

From Prompts to Systems — Volume II

From Prompts to Systems

- Audience

- Introduction

 - What this book is for

 - How this volume is organized

 - Prerequisites and suggested use

- Full text — Parts I through VI

Part I — Mental models and the model lifecycle

- Contents of this part

- Chapter 1 — From playground to product

 - What changes at launch

 - The stack around the model

- Chapter 2 — Training stages in one picture

 - Pretraining

 - Instruction and preference tuning

- Chapter 3 — Choosing a model and reading model cards

 - Deployment constraints

 - Open vs closed weights

- Chapter 4 — Context, memory, and state

 - Truncation and summarization

 - Session state vs retrieval vs fine-tuning

- Try it

Part II — Prompting as engineering

- Contents of this part

- Chapter 1 — Prompt structure and patterns

 - Few-shot and chain-of-thought

- Chapter 2 — Iteration and prompt libraries

 - A/B and offline comparison

 - Templates and guardrails

- Chapter 3 — Failure modes and debugging

 - Techniques

- Chapter 4 — Interaction and UX for LLM features

 - Expectations

 - Team norms

- Try it

Part III — Data, retrieval, and adaptation

- Contents of this part

- Chapter 1 — When to retrieve vs. prompt vs. fine-tune

 - Decision flow (simplified)

- Chapter 2 — Retrieval-augmented generation (RAG)

 - Grounding and “not in corpus”

- Chapter 3 — Curating and labeling data for adaptation

Chapter 4 — Tools and function calling

Orchestration

Try it

Part IV — Evaluation, quality, and safety in practice

Contents of this part

Chapter 1 — What to measure

Human vs model-as-judge vs automation

Chapter 2 — Test sets, regression testing, and CI for LLM features

CI mindset

Chapter 3 — Safety and abuse in product context

Escalation

Try it

Part V — Systems: APIs, deployment, and operations

Contents of this part

Chapter 1 — API design and abstraction layers

Structured outputs

Chapter 2 — Observability and logging

Dashboards

Chapter 3 — Cost, capacity, and rate limits

Chapter 4 — Security basics for LLM applications

Sandboxing

Try it

Part VI — Teams, ethics, and the path forward

Contents of this part

Chapter 1 — Working in cross-functional teams

Documentation

Chapter 2 — Responsible deployment (intermediate stance)

When to escalate

Chapter 3 — Bridge to *From Models to Frontiers*

Skills to sharpen

Try it

Notes

Accessibility

Exercise index (*Try it* sections)

Sample prompts (for APIs and eval)

Glossary export (Volume II core)

Reading list (short)

Optional figures (Mermaid — for HTML/PDF)

Chapter notes

From Prompts to Systems

Intermediate practice with large language models — Volume II

Sharif Uddin

Audience

Developers, analysts, product managers, and technical writers who already use LLMs in daily work and want **clearer mental models, repeatable workflows, and enough depth to ship** features—not just one-off prompts—without yet diving into research-scale training or frontier theory (that is Volume III). You should be comfortable with *From Tokens to Understanding* (Volume I) or equivalent: tokens, context windows, basic prompting, and honest limits of the technology.

Introduction

What this book is for

Volume I answers **what** language models are and **how to begin** using them responsibly. *From Prompts to Systems* answers **how to turn that familiarity into practice you can defend**: choosing approaches (prompt vs. retrieval vs. fine-tuning), **iterating with data, measuring quality, wiring models into software**, and **operating** them with costs, failures, and teams in mind.

The emphasis is **applied**: fewer proofs, more **decision patterns**—when a longer prompt beats a pipeline change, when evaluation should block a release, how to structure an API layer so you can swap models later. The goal is that after this volume you can own an LLM-powered feature from prototype to something your team can maintain—and know what Volume III will unpack if you need to go deeper on training, safety science, or scale.

How this volume is organized

The outline groups material into **pillars** that mirror real projects: **understanding what you are shipping, designing prompts and interactions, adapting behavior** (data, retrieval, light fine-tuning), **evaluating and monitoring, building integrations**, and **shipping responsibly**. Themes like **evaluation** and **safety** appear in more than one part because they are both design-time and runtime concerns.

Chapters are written to support **hands-on progression**: you can draft exercises (small apps, eval sets, prompt libraries) per part even before the prose is final.

Prerequisites and suggested use

You should be comfortable using at least one **chat or API** interface, reading **JSON**-shaped API responses, and thinking in terms of **inputs, outputs, and failure cases**. Light scripting (e.g. Python or JavaScript) will help for later chapters; where code is essential, the book can stay pseudocode-first.

Use the outline as a **manuscript contract**: merge or split chapters as your teaching style evolves. Keep implementation notes, vendor-specific quirks, and bibliography in **Notes** or sidecar files so the main text stays durable.

Full text — Parts I through VI

Part I — Mental models and the model lifecycle

Sharif Uddin

From Prompts to Systems · Volume II

Volume I explained **what** LLMs are and how they behave in the abstract. This part is about **shipping**: the difference between a **demo** and a **product**, how **training stages** show up in model cards, how to **choose** a model, and how **context, memory, and retrieval** fit together before you dive into prompts and RAG in later parts.

Contents of this part

In the full volume table of contents, these correspond to sections 1–4.

	Chapter	What you will take away
1	From playground to product	Latency, cost, ownership, and the stack around the model
2	Training stages in one picture	Pretraining, instruction tuning, preferences—“smart” vs “aligned”
3	Choosing a model and reading model cards	Context, license, deployment; open vs closed weights
4	Context, memory, and state	Truncation, summarization, retrieval vs fine-tune (preview)

Contents (plain list — same as table):

- 1. From playground to product — stack and product concerns.
 - 2. Training stages in one picture — base vs chat, axes of capability.
 - 3. Choosing a model and reading model cards — constraints and cards.
 - 4. Context, memory, and state — what fits, what to design for.
-

Chapter 1 — From playground to product

A **playground** proves the model can do something cool once. A **product** does it **reliably**, under **load**, with **clear ownership** when things break.

What changes at launch

Latency budgets appear: users tolerate seconds for some tasks, milliseconds for others. **Cost** becomes continuous—tokens, seats, or inference hours—not a one-off experiment. **Versioning** matters: which **model ID**, which **prompt template version**, which **retrieval index** went into this answer? **Ownership** means an on-call path: who rolls back a bad prompt change?

The stack around the model

Think in layers: **client** (web, mobile, internal tool) → **application** (your business logic) → **orchestration** (prompt assembly, retrieval, tool routing) → **model provider** (API or self-hosted weights) → **data** (vector store, caches, feature flags). The LLM is one box; **reliability** is a system property.

In this chapter. Shipping adds latency, cost, versioning, and ownership; map the stack before you optimize the model alone.

Chapter 2 — Training stages in one picture

You do not need to train models to **read** how they were produced. Model cards and blog posts refer to **pretraining**, **instruction tuning**, **preference tuning**—different pressures on the same weights.

Pretraining

The model learns broad **language and world patterns** from large text (and often code). That stage sets rough **knowledge cutoff**, fluency, and “raw” behavior—sometimes ill-suited to chat without later steps.

Instruction and preference tuning

Later stages teach **dialogue format**, **refusal** boundaries, and **helpfulness** as judged by raters or automated preference models. **Capability** (reasoning, knowledge) and **alignment** (policy, tone) are related but **not identical**: a base model can be strong yet unsafe for direct users; a chat model can be polite yet shallow.

In this chapter. “Smarter” and “more aligned” are different axes; cards that describe pipeline stages help you predict behavior changes.

Chapter 3 — Choosing a model and reading model cards

Model cards (and API docs) are your **contract** for what to expect: **context length**, **modalities** (text-only vs vision), **languages**, **license** (can you fine-tune? deploy how?), and **known limitations**.

Deployment constraints

Will you run **hosted APIs** only, or **on-prem** / **VPC** with certain certifications? Do you need **structured outputs** or **tool calling** as first-class features? Match **requirements** before you benchmark accuracy.

Open vs closed weights

Open-weight models can be inspected, fine-tuned, and run locally—at the cost of **your** ops burden. **Closed APIs** shift **scaling and safety** to the vendor; you get less control over internals. Neither is universally “better”; fit to **risk**, **budget**, and **latency**.

In this chapter. Read the card for context, license, modalities, and limits before A/B testing clever prompts.

Chapter 4 — Context, memory, and state

Everything the model “remembers” in-session is what you put in the **prompt** (plus optional **retrieved** chunks). There is no hidden long-term memory unless **you** build storage and re-injection.

Truncation and summarization

Long chats and documents exceed **context**. Strategies: **truncate** old turns, **summarize** history, or **start fresh** threads. Each trades **fidelity** for **space**.

Session state vs retrieval vs fine-tuning

- **Session state:** your database of user facts, re-injected when relevant.
- **Retrieval:** fetch **documents** per query (Part III).
- **Fine-tuning:** adjust weights with **curated** examples—use when behavior must be **consistent** and **prompting** + **RAG** are insufficient (details in Volume III).

In this chapter. Context is designed, not magical; retrieval and adaptation are separate tools with separate costs.

Try it

1. **Playground vs product.** List **five** concerns that appear when moving an internal chat demo to a customer-facing feature (e.g. latency, logging). Which are **not** about the raw model quality?

2. **Model card.** Open a **model card** or API doc for a model you might use. Note: **context length**, **license**, **one** limitation, and **one** training stage mentioned.

End of Part I. Previous: From Tokens to Understanding — Volume I · Next: Part II — Prompting as engineering · Or main volume.

Part II — Prompting as engineering

Sharif Uddin

From Prompts to Systems · Volume II

Prompts are not vibes—they are **interfaces**. This part treats them like **versioned artifacts**: structure, iteration, debugging, and **UX** for features that stream, fail, and get reviewed by humans.

Contents of this part

In the full volume table of contents, these correspond to sections 5–8.

	Chapter	What you will take away
1	Prompt structure and patterns	System / user / tools; few-shot and chain-of-thought tradeoffs
2	Iteration and prompt libraries	Versioning, A/B, templates and guardrails
3	Failure modes and debugging	Hallucination, format drift, tool misuse—fixes
4	Interaction and UX for LLM features	Streaming, undo, expectations, team norms

Contents (plain list — same as table):

1. Prompt structure and patterns — roles, patterns, CoT when to use.
2. Iteration and prompt libraries — version control for prompts.
3. Failure modes and debugging — diagnose and constrain.
4. Interaction and UX — streaming, expectations, collaboration.

Chapter 1 — Prompt structure and patterns

System messages set global rules; **user** messages carry the task; **assistant** history is prior output; **tool** messages carry **results** from your code. APIs differ in naming—map them mentally to those roles.

Few-shot and chain-of-thought

Few-shot examples teach **format** faster than prose. **Chain-of-thought** (“think step by step”) can improve **reasoning-heavy** tasks—and can **hurt** when you need **speed**, **brevity**, or when the model **leaks** reasoning you do not want users to see. **Hide** chain-of-thought in internal prompts when appropriate.

In this chapter. Roles are contracts; few-shot locks format; CoT is a tool with tradeoffs.

Chapter 2 — Iteration and prompt libraries

Treat prompts like **code**: **branch**, **review**, and **tag** versions (e.g. `prompt-v1.3`, `holiday-tone`). Store them in **git** or a **CMS**—not only in a chat box.

A/B and offline comparison

For changes that affect **metrics**, run **offline** evals on a **golden set** (Part IV) before live A/B. **Online** experiments need **traffic** and **guardrails**—do not ship prompt changes without a rollback path.

Templates and guardrails

Variables (user name, locale) belong in templates, not copy-paste. **Guardrails** (“never output raw JSON to end users”) belong in **system** or **post-processing**, not hope.

In this chapter. Version prompts, measure before wide rollout, separate data from policy.

Chapter 3 — Failure modes and debugging

Hallucination and **format errors** are not rare bugs—they are **baseline risks**. **Sycophancy** (agreeing with the user) can break **safety** or **accuracy**.

Techniques

Decompose tasks into steps; **self-check** (“list assumptions before answering”); **constrain** output (**JSON schema**, **regex** validation). For **tools**, validate arguments in **code** before execution.

In this chapter. Debug with structure: smaller steps, validation, and explicit uncertainty.

Chapter 4 — Interaction and UX for LLM features

Streaming tokens improves perceived speed; pair with **cancel** and **retry**. **Undo** or **edit** last turn reduces frustration when the model drifts.

Expectations

Tell users **what the feature does not do** (no legal advice, no real-time data unless wired). **Loading** and **error** states should be honest—masking failures erodes trust.

Team norms

Decide how **engineering** and **design** review **tone**, **disclosure** of AI, and **fallbacks** when the model is down. Document **prompts** and **model IDs** in **runbooks**.

In this chapter. UX and ops are part of the feature—not polish after the fact.

Try it

1. **Template.** Write a **system** prompt + one **user** template with `{variable}` slots for a task you care about. Version it (`v0.1`) in a file.
2. **Debug.** Reproduce **one** failure (wrong format or hallucination). Change **only** the prompt: add a constraint or a single-shot example. Did behavior improve?

End of Part II. Previous: Part I — Mental models and the model lifecycle · Next: Part III — Data, retrieval, and adaptation · Or main volume.

Part III — Data, retrieval, and adaptation

Sharif Uddin

From Prompts to Systems · Volume II

Not every problem is a **prompt** problem. This part is about **when to fetch facts**, **how RAG fits together** at a practical level, **data for adaptation**, and **tools** that connect models to your systems **safely**.

Contents of this part

In the full volume table of contents, these correspond to sections 9–12.

	Chapter	What you will take away
1	When to retrieve vs. prompt vs. fine-tune	Decision flow: freshness, privacy, cost
2	Retrieval-augmented generation (RAG)	Chunking, embeddings, grounding, citations
3	Curating and labeling data for adaptation	Quality, formats, synthetic data risks
4	Tools and function calling	Safe tools, retries, orchestration patterns

Contents (plain list — same as table):

1. When to retrieve vs. prompt vs. fine-tune — decision flow.
2. RAG — chunk, embed, retrieve, generate.
3. Curating data — labels, quality, synthetic caveats.
4. Tools and function calling — contracts and orchestration.

Chapter 1 — When to retrieve vs. prompt vs. fine-tune

Prompting alone works when **knowledge** is in the model or **not critical**. **Retrieval** when facts are **fresh, private**, or **too large** to memorize in weights. **Fine-tuning** when you need **stable style or behavior** across many inputs and prompts are brittle—subject to data cost and Volume III depth.

Decision flow (simplified)

- Need **latest** docs or **internal** wiki? → **RAG** (or live API tools).
- Need **consistent brand voice** on millions of requests? → consider **fine-tuning** + eval.
- One-off **format** or **tone**? → **prompt** + few-shot first.

In this chapter. Match mechanism to constraint: freshness, privacy, cost, and behavior stability.

Chapter 2 — Retrieval-augmented generation (RAG)

RAG = **retrieve** relevant chunks, **inject** into context, **generate**. Quality hinges on **chunking** (size, overlap), **embedding** model choice, and **index** freshness.

Grounding and “not in corpus”

Instruct the model to **cite** or **quote** retrieved text; **refuse** when chunks do not support an answer. **Hallucination** on top of bad retrieval is **worse** than admitting ignorance—design **fallback** UX.

In this chapter. RAG is a pipeline: bad chunks in → fluent garbage out; invest in retrieval quality.

Chapter 3 — Curating and labeling data for adaptation

Whether you **fine-tune** or **build eval sets**, **data quality** beats raw size. **Duplicates** and **biased** raters distort behavior. **Synthetic** data can bootstrap—but may **amplify** model biases; **human** spot checks matter.

In this chapter. Label carefully; watch for feedback loops; synthetic data is a lever, not a cure-all.

Chapter 4 — Tools and function calling

Tools are **your** functions: query DB, create ticket, fetch URL—with **schemas** the model fills. **Never** give the model **open shell** or **arbitrary** network unless you **sandbox**.

Orchestration

Patterns: **single** model with tools; **router** model picks a skill; **small pipeline** (classify → retrieve → answer). Pick **complexity** to match **failure modes** you can test.

In this chapter. Tools need schemas, validation, retries, and least privilege—same as any integration.

Try it

1. **Decision.** Pick one real task (e.g. “answer from our handbook”). Write **two sentences**: why **RAG** vs **prompt-only** for that task.
 2. **Tool contract.** Sketch a **JSON schema** for one function (name + two parameters). What could go wrong if the model **hallucinates** an argument?
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End of Part III. Previous: Part II — Prompting as engineering · Next: Part IV — Evaluation, quality, and safety in practice · Or main volume.

Part IV — Evaluation, quality, and safety in practice

Sharif Uddin

From Prompts to Systems · Volume II

You ship what you measure. This part covers **metrics**, **golden sets** and **regression** discipline, and **product safety** layers—so improvements do not become accidents at scale.

Contents of this part

In the full volume table of contents, these correspond to sections 13–15.

	Chapter	What you will take away
1	What to measure	Task quality, latency, cost, human vs automatic judgment
2	Test sets, regression testing, and CI	Golden sets, model upgrades, breaking changes
3	Safety and abuse in product context	Policies, classifiers, PII, escalation

Contents (plain list — same as table):

- 1. What to measure — metrics that match the job.
 - 2. Test sets and regression — golden sets, CI mindset.
 - 3. Safety and abuse — layers and data handling.
-

Chapter 1 — What to measure

Accuracy on a task is not one number: **correctness** of facts, **helpfulness**, **format** adherence, **latency**, **cost per success**, and **harm** rate. Pick **few** primary metrics aligned with **user value**—not every leaderboard score.

Human vs model-as-judge vs automation

Human labels are gold but expensive. **Model-as-judge** scales but inherits **biases**. **Automatic** checks (JSON valid, regex, unit tests on tool args) are **cheap**—combine layers.

In this chapter. Define success in user terms; mix human spot checks with scalable signals.

Chapter 2 — Test sets, regression testing, and CI for LLM features

A **golden set** is a **fixed** batch of inputs (and often expected properties) you run on **every** prompt or model change. **Snapshot** outputs or **scores**—watch for **drift** when providers **silent-upgrade** models.

CI mindset

Block releases when golden metrics fall below threshold—same as unit tests. **Pin** model versions in config until you **re-validate**.

In this chapter. Treat prompt and model changes like code changes: test before merge.

Chapter 3 — Safety and abuse in product context

Policy layers (blocklists, classifiers, moderation APIs) sit **beside** the model—not as a substitute for **good prompts**, but as **defense in depth**. **PII: minimize** what you log; **retention** policies are part of **security**.

Escalation

Define **when** a human reviews **edge cases**—legal, self-harm, targeted harassment. **Automation** should not be the last word on every abuse report.

In this chapter. Safety is product design: layers, logging discipline, and human escalation paths.

Try it

1. **One metric.** Pick a feature you know. Name **one** measurable outcome (e.g. “% answers with valid JSON”) and **one** human-judged aspect.
 2. **Golden test.** Write **three** input prompts you would put in a **regression** suite for that feature and **what** you would check automatically vs manually.
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End of Part IV. Previous: Part III — Data, retrieval, and adaptation · Next: Part V — Systems: APIs, deployment, and operations · Or main volume.

Part V — Systems: APIs, deployment, and operations

Sharif Uddin

From Prompts to Systems · Volume II

Models live behind **APIs**. This part covers **wrappers**, **observability**, **cost and rate limits**, and **security** basics—so production incidents are **boring**, not mysterious.

Contents of this part

In the full volume table of contents, these correspond to sections 16–19.

	Chapter	What you will take away
1	API design and abstraction layers	Retries, streaming, structured outputs
2	Observability and logging	Traces, redaction, dashboards
3	Cost, capacity, and rate limits	Tokens, caching, right-sizing
4	Security basics for LLM applications	Prompt injection, trust boundaries, sandboxing

Contents (plain list — same as table):

- 1. API design — abstraction, timeouts, streaming.
- 2. Observability — logs, traces, what not to log.
- 3. Cost and capacity — tokens, caching, limits.
- 4. Security — injection, tools, least privilege.

Chapter 1 — API design and abstraction layers

Wrap vendor APIs behind **your** interface: **model name**, **temperature**, **max tokens** as **config**, not scattered strings. **Timeouts** and **retries** with **idempotency** keys for **side-effecting** tool calls.

Structured outputs

Parse **JSON** defensively—models **stray**. **Schema validation** before downstream use. **Streaming** partial tokens: buffer until **valid** chunk if needed.

In this chapter. One abstraction layer makes swaps and tests easier; validate all structured output.

Chapter 2 — Observability and logging

Log **request IDs**, **latency**, **token counts**, **model ID**, **error codes**—not **raw** user PII unless required. **Trace** multi-step flows (retrieve → generate → tool).

Dashboards

Track **error rate**, **p95 latency**, **cost per request**, and **quality** proxies from **offline** evals. **Alert** on spikes—often the first sign of a **bad deploy** or **provider** issue.

In this chapter. Observability enables blameless postmortems; redact before you aggregate.

Chapter 3 — Cost, capacity, and rate limits

Token accounting per **route** and **customer**. **Cache** repeated **context** where safe. **Batch** where latency allows. **Right-size** models: small model for **classification**, large for **generation**—if routing is worth the complexity.

In this chapter. Cost is a feature; measure before you optimize the wrong layer.

Chapter 4 — Security basics for LLM applications

Prompt injection: untrusted text **in** the prompt **directs** the model to ignore instructions or **exfiltrate** data. **Mitigations:** **separate** trusted vs untrusted blocks, **downgrade** privileges for tool calls, **human** approval for **irreversible** actions.

Sandboxing

Run tools in **minimal** environments; **no** raw SQL from model output without **validation**. **Trust boundaries** mirror classic **web security**—with new angles.

In this chapter. Treat the model as an untrusted client; tools are the real power.

Try it

1. **Envelope.** Sketch a **pseudo-request** object your API would send upstream: fields for **model**, **messages**, **tools**, **timeout**. What is **one** field you would **not** expose to the browser?
2. **Log policy.** List **three** things worth logging for a single LLM request and **one** thing you would **strip** or **hash** by default.

End of Part V. Previous: Part IV — Evaluation, quality, and safety in practice · Next: Part VI — Teams, ethics, and the path forward · Or main volume.

Part VI — Teams, ethics, and the path forward

Sharif Uddin

From Prompts to Systems · Volume II

Shipping LLM features is **cross-functional**. This part covers **roles**, **documentation**, **responsible deployment** at an intermediate level, and the **bridge to Volume III** when you need research depth.

Contents of this part

In the full volume table of contents, these correspond to sections 20–22.

	Chapter	What you will take away
1	Working in cross-functional teams	ML, backend, design, legal; handoffs
2	Responsible deployment (intermediate stance)	Transparency, escalation, proportionality
3	Bridge to <i>From Models to Frontiers</i>	What Volume III adds; skills to sharpen

Contents (plain list — same as table):

- 1. Working in cross-functional teams — roles and checkpoints.
- 2. Responsible deployment — transparency without Volume III depth.
- 3. Bridge to Volume III — frontier topics and reading path.

Chapter 1 — Working in cross-functional teams

Product defines success; **design** shapes trust and disclosure; **backend** owns latency and **API** contracts; **ML** (if present) owns **eval** and **model** choice; **legal** reviews **data** and **claims**. **No** single role “owns” safety—**checkpoints** (design review, legal sign-off for risky domains) prevent late surprises.

Documentation

Runbooks: which **model**, which **prompt version**, how to **rollback**. **Playbooks** for **incidents** (spike in abuse, provider outage). Handoff should not depend on **one** engineer’s head.

In this chapter. Clarity of ownership beats heroics; document what you ship.

Chapter 2 — Responsible deployment (intermediate stance)

Transparency: users should know **when** AI is involved and **how to** escalate or opt out where feasible. **User control** over **data** and **settings** reduces harm and **trust** erosion. **Proportionality**: not every feature needs the **largest** model—**cost** and **risk** scale together.

When to escalate

Bring **safety specialists** or **policy** for **high-stakes** domains, **jurisdictional** questions, or **public** commitments. Volume III goes deeper on **alignment science** and **governance**; here, know **when** to ask for help.

In this chapter. Ship with clear disclosure, controls, and escalation paths—not only metrics.

Chapter 3 — Bridge to *From Models to Frontiers*

Volume III, *From Models to Frontiers*, is for **depth: scaling laws, training stacks, alignment** research, **efficiency**, **multimodal** and **agentic** systems, and **open problems**. Read it when you **build** or **evaluate** at the frontier—not when you only **consume** APIs.

Skills to sharpen

- **Reading papers** and **model cards** critically.
- **Basic** distributed training / inference vocabulary (even if you do not train).
- **Structured** thinking about **failure modes** and **societal** impact.

In this chapter. Volume II gets you to **ship**; Volume III helps you **reason** about what is coming next.

Try it

1. **RACI-style.** For one LLM feature, name **who** is accountable for: **model choice, prompt changes, customer comms** if something goes wrong.
 2. **Volume III preview.** Open *From Models to Frontiers*; read the **introduction** outline. **One** topic you want to learn next—write it down.
-

End of Part VI — Volume II. Previous: Part V — Systems: APIs, deployment, and operations · Next: From Models to Frontiers — Volume III · Or main volume.

Notes

This section collects **optional material** for Volume II: sample prompts, a reading list, a **glossary export**, an **exercise index**, optional **figures**, and accessibility notes. It does not replace the parts.

Accessibility

Part files use **tables** for “Contents of this part.” Each part also includes a **plain list** mirroring the table for readers or tools that do not render tables well.

Exercise index (*Try it* sections)

Part	File	Rough focus
I	part-i	Playground vs product; read a model card
II	part-ii	Prompt template; debug one failure
III	part-iii	RAG vs prompt decision; tool contract
IV	part-iv	Define one metric; golden test idea
V	part-v	Sketch an API envelope; what to log
VI	part-vi	RACI snippet; Volume III preview

Exercise index (plain list): Part I — product vs demo, model card · Part II — template + debug · Part III — RAG vs prompt, tools · Part IV — metric, golden tests · Part V — API, observability · Part VI — team handoff, frontier preview.

Sample prompts (for APIs and eval)

1. **System prompt skeleton.** “You are [role]. Reply in [format]. If unsure, say what is missing. Never invent [constraint].”
2. **Eval rubric.** “Score the assistant answer 1–5 on correctness, helpfulness, and safety. Give one sentence of justification each.”
3. **RAG grounding.** “Using only the provided CONTEXT blocks, answer the question. If CONTEXT is insufficient, say so.”
4. **Regression check.** “Given INPUT and EXPECTED_SHAPE, does OUTPUT parse as valid JSON with keys [...]?”

Glossary export (Volume II core)

- **RAG (retrieval-augmented generation)** — Fetching relevant documents (or chunks) into **context** before generation, so answers can **ground** in supplied text.
- **Golden set** — A fixed **evaluation** set of inputs (and often reference outputs) used to detect **regressions** when models or prompts change.
- **Model card** — A structured description of a model’s **intent**, **data**, **limitations**, and **evaluation**—read before you commit.
- **Tool / function calling** — The model emits **structured calls** (e.g. JSON) to bounded functions your app implements; not arbitrary code execution.
- **Prompt injection** — User or untrusted content that **manipulates** the model into bypassing instructions or **misusing** tools—**trust boundaries** matter.

Reading list (short)

- Provider **API documentation** and **safety** guides for the stack you ship.
- Papers or posts on **RAG** architecture and **eval harnesses** once you need depth—*From Models to Frontiers* points to research-scale material.

- *From Tokens to Understanding* (Volume I) for vocabulary; *From Models to Frontiers* (Volume III) for training, scale, and frontier topics.

Optional figures (Mermaid — for HTML/PDF)

Stack: app → orchestration → model

```
flowchart TB
    UI[Application / UI] --> ORCH[Orchestration]
    ORCH --> API[Model API]
    ORCH --> RET[Retrieval / tools]
    RET --> STORE[(Data stores)]
```

Eval loop

```
flowchart LR
    P[Prompt + model] --> O[Output]
    O --> M["Metrics / human review"]
    M -->|fail| P
    M -->|pass| Ship["Ship / monitor"]
```

Chapter notes

Add your own manuscript notes, ticket links, and per-environment quirks below.